Def. Let G be a group, and $H \leq G$ be a subgroup.
Given an element $g \in G$, define left-coset of H as

$$gH \overset{def}{=} \{gh \mid h \in H\}.$$

Lemma. Let G be a group, and $H \leq G$ be a subgroup. Then, for each $g_1, g_2 \in G$,

$g_1 H = g_2 H$ if and only if $g_1^{-1} g_2 \in H$.

Proof. Given $g_1, g_2 \in G$, suppose that $g_1 H = g_2 H$. Then, since $g_2 \in g_2 H = g_1 H$ ($1 \in H$),

$g_2 = g_1 h$ for some $h \in H$, thus $g_1^{-1} g_2 \in H$.

Conversely, let $g_1, g_2 \in G$ be given s.t. $g_1^{-1} g_2 \in H$. That is, $g_1 h = g_2$ for some $h \in H$.
Given $g_1 h_1 \in g_1 H$, $g_1 h_1 = g_2 h^{-1} h_1 \in g_2 H$. Given $g_2 h_2 \in g_2 H$, $g_2 h_2 = g_1 h h_2 \in g_1 H$ ☒

Q. In left coset sence, $g_1 H = g_2 H \iff g_1 g_2^{-1} \in H$ hold?
A. No!, S_3, $H = \langle (1,2) \rangle$, $g_1 = (1\ 3)$ and $g_2 = (1\ 2\ 3)$

Thm. Lagrange Thm
If H is a subgroup of a group G, then $|G|/|H| = [G:H]$
Prof. Given a subgroup $H \leq G$, define a relation on G as;

$$x \sim y \iff xH = yH$$

Then, it is an equivalence relation, so a collection $\{gH \mid g \in G\}$ forms a partition.
Moreover, for each $g \in G$, a map

$$H \to gH \,;\, h \mapsto gh$$

is onto by definition and one to one by the cancelation law, so $|H| = |gH|$.

┌ Isomorphism Theorem.

┌ 1^{st} Isomorphism theorem.

For any Homomorphism $\varphi: G \to G'$,

$$G/\ker\varphi \cong \operatorname{Im}\varphi.$$

+ If G is finite, then $|G| = |\ker\varphi| \cdot |\operatorname{Im}\varphi|$.

prof. Since $\ker\varphi \trianglelefteq G$, $G/\ker\varphi$ is well-defined as a group. Define

$$\tilde{\varphi}: G/\ker\varphi \to \operatorname{Im}\varphi : g\ker\varphi \mapsto \varphi(g).$$

Then, $g_1\ker\varphi = g_2\ker\varphi \iff g_1 g_2^{-1} \in \ker\varphi \iff \varphi(g_1) = \varphi(g_2)$

so $\tilde{\varphi}$ is well-defined and injective, and the surjective is clear. ∎

Note: the First Isomorphism theorem gives:

$N \trianglelefteq G$ is Normal subgroup if and only if $\pi: G \to G/N$ is Homomorphism, $\ker\pi = N$.

┌ Prop. Let $\varphi: G \to G'$ be a Homomorphism.

1). $H \leq G \Rightarrow \varphi[H] \leq G'$
2). $N \trianglelefteq G \Rightarrow \varphi[N] \trianglelefteq \varphi[G] = \operatorname{Im}\varphi$.
3). $H' \leq G' \Rightarrow \varphi^{-1}[H'] \leq G$
4). $N' \trianglelefteq G' \Rightarrow \varphi^{-1}[N'] \trianglelefteq G$.

「2^{nd} Isomorphism theorem.

Let G be a group, and let A, B be subgroups of G satisfying $A \leq N_G(B)$.
Then, $AB \leq G$ and $B \trianglelefteq AB$, $A \cap B \trianglelefteq A$. Further,

$$AB/B \cong A/A\cap B$$

Proof. Since A normilizes B,

$$AB = \bigcup_{a\in A} aB = \bigcup_{a\in A} Ba = BA$$

and so given $a_1b_1, a_2b_2 \in AB$, $a_1b_1(a_2b_2)^{-1} = a_1b_1b_2^{-1}a_2^{-1} \in AB$, so $AB \leq G$.
$B \leq AB$ is trivial, and given $ab \in AB$ and $g \in B$, $abgb^{-1}a^{-1} \in B$, so $B \trianglelefteq AB$.
$A\cap B \trianglelefteq A$ is also trivial (These are given by the fact that A nomalizes B).
Now, define a map

$$\varphi : A \to AB/B : a \mapsto aB.$$

It is well-defined since $A \subseteq AB$ and it is Homomorphism because: given $a_1, a_2 \in A$,

$$\varphi(a_1a_2) = a_1a_2B = a_1B \cdot a_2B$$

and onto: given $abB \in AB/B$, take $a \in A$, then $\varphi(a) = aB = abB$.
And, the kernel is $\operatorname{Ker}\varphi = \{g \in A : gB = B\} = \{g \in A \mid g \in B\} = A \cap B$. Consequently, the first isomorphism theorem gives $A/A\cap B \cong AB/B$. $\blacksquare$

「3^{rd} Isomorphism Theorem.

Let G be a group and $H, K \trianglelefteq G$ be normal subgroups with $H \leq K$.
Then, $K/H \trianglelefteq G/H$ and $[G/H]/[K/H] \cong G/K$.

proof. Since $K \trianglelefteq G$, $K/H \trianglelefteq G/H$ directly. Define a map

$$\varphi : G/H \to G/K : gH \mapsto gK.$$

It is well-defined, because: given $g_1, g_2 \in G$,

$$g_1H = g_2H \Leftrightarrow g_1g_2^{-1} \in H \underset{H \leq K}{\Rightarrow} g_1g_2^{-1} \in K \Leftrightarrow g_1K = g_2K = \varphi(g_1) = \varphi(g_2)$$

(so, injectiveness is given only $H = K$).

Surjectivity and Homomorphism are clear. Moreover,

$$\text{Ker}\,\varphi = \{gH \in G/H \mid gK = K\} = \{gH \in G/H \mid g \in K\} = K/H.$$

Finally, the 1^{st} isomorphism theorem gives the result.

4th Isomorphism Theorem.

Homomorphism.

Def. Let G, H be groups. A map $\varphi: G \to H$ is called a Homomorphism if for all $g_1, g_2 \in G$, $\varphi(g_1 \cdot_G g_2) = \varphi(g_1) \cdot_H \varphi(g_2)$.

Basic Properties.

$\varphi(g^n) = \varphi(g)^n$ for any $g \in G$ and $n \in \mathbb{Z}$. So, $|\varphi(g)| \mid |g|$.